

A New Approach for Performance Evaluation of Survivability Services in All-Optical Networks

André Soares, William Giazza, and Paulo Cunha.

Abstract— We propose in this paper a new set of measures (blocking probability, restorability, blocking probability per source-destination pair, blocking probability due to the absence of resources in the backup route, etc.) for performance evaluation of survivability services in all-optical networks. A comparison between the Two Step with Backtracking (TSB) and the Simple Restoration (SR), two important references of all-optical network survivability schemes is made using such set of measures. We also propose a new technique to provide survivability for all-optical networks, named Adaptive Survivability (ADPS). Besides being more flexible, we show that our scheme has a more balanced performance in terms of blocking probability and restorability when compared to the Two Step with Backtracking and the Simple Restoration techniques.

Index Terms— Survivability Optical Networks, WDM Network.

I. INTRODUCTION

Wavelength Division Multiplexing (WDM) is a technique which improves significantly the use of optical fiber bandwidth. By applying such technique, several optical channels can be established at the same time, working at different wavelengths in a single optical fiber [1]. In a wavelength-routed WDM network, end users are able to communicate with one another via all-optical WDM channels (lightpaths). In those optical networks, data are routed from source node to destination node through one or more wavelengths associated with one lightpath. Each lightpath may easily support transmission rates up to 40 Gbps with available commercial technology.

Once a lightpath is established it uses the full bandwidth of each wavelength assigned to it until the connection is tear down. When a connection is finished all wavelengths having been used by become available again and can be used for future connections. To establish a lightpath it is necessary to choose a route and to assign a wavelength (or wavelengths) at the different links forming the specified route. This problem is known as Routing and Wavelength Assignment (RWA) problem. In an optical network with dynamic traffic, connections arrive upon demand. For optimization purposes of

an all-optical network in terms of resource utilization, routing and wavelength assignment have to be planned with the aim of minimizing the connection blocking probability. Route and wavelength choices need to be done quickly on-line [2]. Without wavelength conversion capability at intermediaries nodes, a lightpath must occupy the same wavelength on all the links of the established route. This property is known as wavelength continuity constraint.

Wavelength converters are devices that can convert an input wavelength in a different output wavelength. A network with full wavelength conversion capability works as a common circuit-switched telephony network. Nevertheless, price and relative immaturity of the wavelength converter technology still limit its use in all-optical network infrastructures [3].

In a wavelength-routed WDM network, a node or link failure affects all the lightpaths that traverse the failed components. Links in optical WDM networks are supposed to have high transmission capacity, therefore a link failure may represent the loss of a large amount of data. Besides, applications using all-optical network services are usually highly sensitive to delays and must be in operation 24 hours a day, requiring high availability of services [4,5]. Therefore, a failure in such kind of network may imply in loss of large amounts of data and interruption of critical services. To face these situations, an all-optical network must implement mechanisms to guarantee that failures can be quickly and efficiently recovered.

The ability of an optical network to survive after the occurrence of failures is known as optical network survivability. Optical network survivability may be achieved with the implementation of protection and/or restoration mechanisms.

In this work we propose the use of a new set of measures for performance evaluation of survivability services in all-optical networks. Applying a more complete set of measures, one can have a more detailed view of each network survivability strategy. This approach allows for example to localize bottlenecks or unbalanced configurations. We also propose in this work a new technique to provide survivability for all-optical networks, named Adaptive Survivability.

The rest of this paper is organized as follows: in Section II, we present some important concepts of survivability in optical networking. Related works and our contribution are presented in Section III. In Section IV a performance evaluation study comparing the Two Step with Backtracking (TSB) and the Simple Restoration (SR) survivability techniques is made. The

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Adaptive Survivability (ADPS) technique is presented and evaluated in Section V. Finally, Section VI presents some conclusions.

II. SURVIVABILITY IN ALL-OPTICAL NETWORK

Optical survivability is typically classified in protection or restoration schemes. Protection techniques consist of reserving redundant resources (i.e. before the failure happens) that on an eventual failure will replace the failed ones. Alternatively, in restoration schemes, redundant resources are not allocated. Restoration tries to find disjoint resources in order to recover the failure. Protection and restoration schemes can be based in the path, in the link or in the sub-path domains. Detailed classification of protection and restoration schemes are found in [4,5].

In path protection two disjoint routes (primary and backup) are assigned for each connection. The former is called primary route, and it is used in the absence of failure. The latter, backup route, is used if any resources of the primary route fails. In the link protection one backup route is associated for each link of primary route. When applying path protection, the time needed to recover a connection using the backup route is usually higher than the link protection option. However, the link protection reserves a larger amount of resources, since there is a backup route for each link of the primary route. In the sub-path protection, only the sub-paths with k links are protected. The sub-path protection tries to balance the pros and cons of the path and link protection in an attempt to equalize the compromise between recovery time and number of resources previously reserved.

On average, one fiber failure occurs in a rate of 4.39 for each 1609 km in a year. The average time to recover a fiber failure is 12 hours, which is much less than the time between the occurrences of two failures [4]. Therefore, the probability of two or more failures happen at the same time is low. Such single failure model, where more than one failure never occurs at the same time, is used in major works related in the literature [6, 7]. Under this assumption (single failure model), it is possible to guarantee 100% of survivability using two disjoint routes for each lightpath. In such case, the protection mechanism can also be classified as *dedicated protection* and *shared protection*, according to the backup route resource sharing capacity. In dedicated protection, backup route resources are used to protect one, and only one, lightpath. The shared protection techniques try to share the resources of the backup between backup routes that have disjoint primary routes. Taking into account the single failure model, only one of two connections with disjoint primary routes can be interrupted by a failure occurrence. Their backup resources may share the same resources. This alternative aims to considerably reduce the number of resources needed to provide optical networks survivability.

Restoration schemes do not reserve redundant resources, but try to reallocate the failed route. The use of restoration

schemes does not guarantee that a lightpath will survive in the case of failure. The restoration routes can be pre-planned or dynamically computed (on-line). In the former procedure, a possible restoration route can be pre-computed for each connection. In the latter, the restoration route is computed after (on-line) detecting failure using the current network state.

III. RELATED WORK AND OUR CONTRIBUTION

Extensive research has been done in the area of network failure protection and restoration [4-8]. A review of the main techniques to provide survivability in all-optical networks is presented in [4]. In [5] Zang and Mukherjee studied all-optical networks without wavelength conversion under dynamic traffic. This work compares dedicated-path protection and shared-path protection using blocking probability, connection setup delay, restoration time and data loss per cut as performance measures. In [6], Zhang *et. al* make a comparison between one-step and two-step protection approaches to compute disjoint routes (primary and backup path) in an optical network with and without wavelength conversion, only using blocking probability as measure. The work in [7] proposes CAFES (Compute a Feasible Solution) and OPT (Optimization) to compute shared path protection in an optical network with wavelength conversion. Here, the simulation results are obtained using blocking probability, average distance hop and resource overbuild as measures. In [8] the authors analyze the active restoration scheme in terms of blocking probability and restorability.

In all these previous works [5-8] and in the majority of the studies found, comparisons are made and the new techniques are proposed without considering some important measures, as for example network utilization and blocking probability per source-destination pair.

As far as we know, no work has ever been done comparing protection and restoration schemes by analyzing the following measures together:

1. Blocking probability;
2. Blocking probability for each pair (source, destination);
3. Blocking probability due to absence of resource in the backup route;
4. Restorability;
5. Average network utilization and
6. Restoration time.

To make this comparison better detailed, we have chosen one scheme of protection and other of restoration. For the pre-planned protection we have used the **Two Step with Backtracking (TSB)** that computes, in the first step, the shortest route for the primary route. In the second step it computes the shortest disjoint route of the primary route for the backup route. The backtracking procedure prevents that the primary route allocation makes it impossible to find a backup route [6].

Figure 1 shows a trap topology when it is required to compute two disjoint path routes. Applying the two step without Backtracking approach, for example, to try to attend the lightpath request (2,9), choosing the primary route {2, 3, 4,

10, 9} makes the backup route finding impossible, as shown in Figures 1a and 1b.

In this work we assume that TSB will be used to implement a 1:1 protection. In that case, the redundant resource can be used to attend traffic without survivability requirements. Such advantage is achieved with the sacrifice of restoration time (time necessary to repair the failure).

Simple Restoration (SR) scheme establishes the lightpath using fixed route with shortest path and pre-computes the restoration route disjoint of the primary route. The simple restoration also implements the backtracking. For the restoration schemes, we also use the term “primary route” to make reference to the route used before failure occurrence.

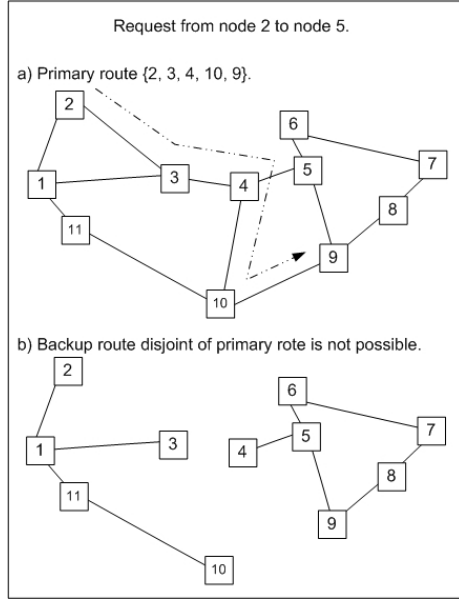


Fig. 1. Example of a trap topology.

Sometimes, when making reference to restoration techniques, we use backup and restoration like synonymous terms. The schemes TSB and SR are chosen in this study due to control plane low complexity purposes.

It is important to note that the restoration time achieved with TSB and SR is similar because both schemes use the same backup route and the SR pre-computes the restoration routes. The time needed to apply the survivability techniques is: 1) time to detect the failure; 2) time for signaling each node of the backup route; and 3) time to reconfigure the OXC's nodes (optical cross-connects).

A. Adaptive Survivability scheme ADPS

From the analysis of the set of measures proposed in this paper we conceive a new strategy for survivability in all-optical networks called **Adaptive Survivability (ADPS)**. The main characteristic of ADPS is to achieve a balanced performance between the protection (TSB) and the restoration (SR) schemes.

According to the results obtained using measures such as blocking probability, network utilization and restorability, we

observe that SR achieves better performance than TSB under low levels of network utilization. However, under high levels of network utilization, TSB presents better results than SR. The basic idea supporting the Adaptive Survivability scheme is to have an scheme which operates as SR under low network utilization and as TSB under high levels of network utilization. This adaptive behavior is implemented according to an adaptive parameter C . If the network utilization is larger then the adaptive parameter C , TSB is used. Otherwise SR is applied.

Despite Adaptive Survivability (ADPS) is carried out applying *Two Step with Backtracking* and *Simple Restoration* as references, it may be applied to make performance adaptation between others protection and restoration schemes.

IV. PERFORMANCE EVALUATION

Due to the complexity of the analytical solution for all measures and given the unavailability of real systems we use simulation techniques in this work. The range of our simulation tool, the SimRWA [9], was extended to support these new functions. Such tool has been partly validated through simulation results and analytical models presented in [10-12]. In the Figure 2, one simulation result of the SimRWA is shown. The results of the SimRWA are compared with other simulation results reported in [10] regard blocking probability under NSF network topology (16-nodes) applying path protection. Fig. 3 shows an example of SimRWA screenshot.

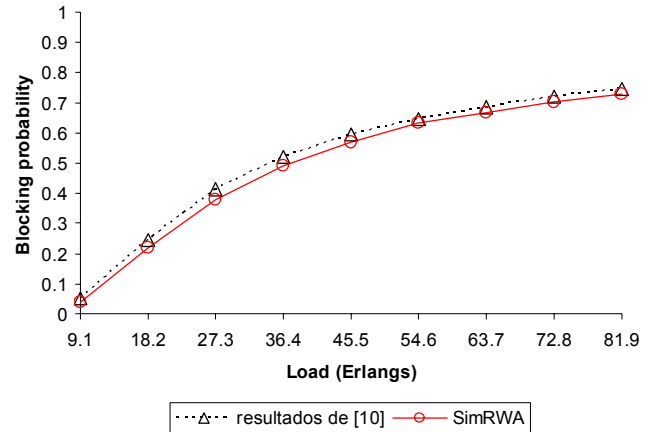


Fig. 2. Validation of SimRWA results.

A dynamic traffic model is used in the simulations, where connection requests arrive at the network following a Poisson process with an average rate λ . The load is distributed uniformly for all nodes. The holding time of the connections is exponentially distributed with an average equals to $1/\mu$. The Abilene and EON network topologies (Figures 4a and 4b) have been used in our simulations. Every link is bi-directional having 40 wavelengths on both ways. The network load is $\rho = \lambda/\mu$. The wavelength assignment algorithm used is First-Fit (FF).

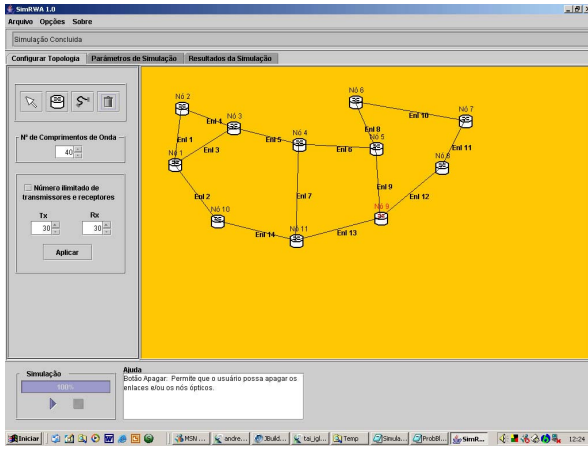


Fig. 3. Example of a SimRWA screenshot.

For each simulation, 10 replications have been made with different seeds of random variable, and for each replication 100,000 connections requisitions have been generated. All results show the confidence intervals calculated with 95% of confidence level.

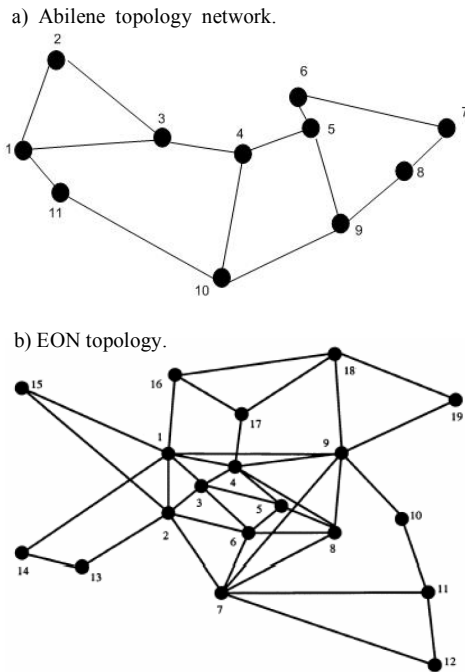


Fig. 4. Network topologies used in our simulations.

Generation of link failures also follows a Poisson process. The average link failure rate is λ_f and the average time to fix the link failure is exponentially distributed with an average $1/\mu_f$. The failures are uniformly distributed for all links. The failure of one link affects both directions, that is, one failure on the link $\langle a, b \rangle$ also affects the link $\langle b, a \rangle$. We have used the failure occurrence statistics presented in [4-7]. On average, one fiber failure occurs in a rate of 4.39 for each 1609 km in a year. The average time to recover a fiber failure is 12 hours. In order to simulate a sufficient number of link failures in our

system, we have had to increase the failure rate and shrink the failure-fixing delay. This practice is also used in [5].

Blocking probability measure is computed by dividing the number of blocked connections by the number of connections requested. When a link failure occurs the restorability measurement is calculated by dividing the number of both recovered and affected connections respectively.

Our results used for comparing TSB, SR and ADPS (C=50%) schemes are shown in Figures 5, 6, 7 and 8 respectively.

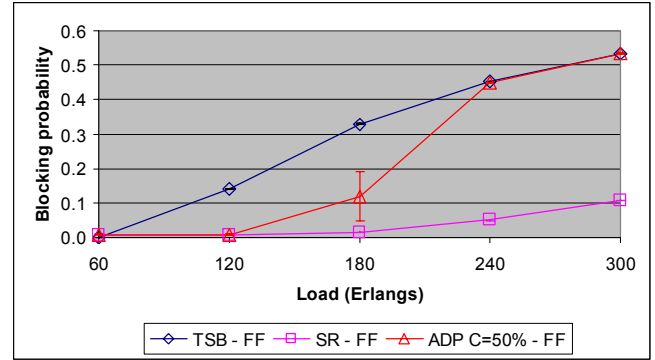


Fig. 5. Blocking probability.

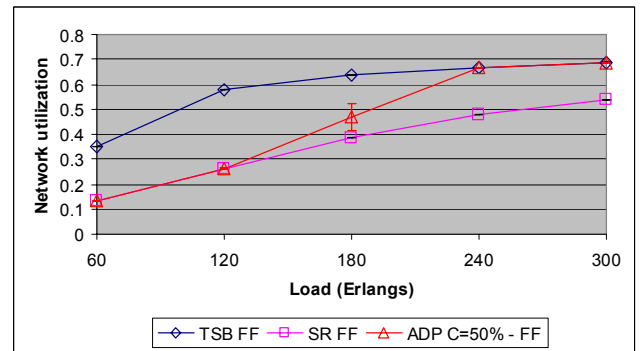


Fig. 6. Average network utilization.

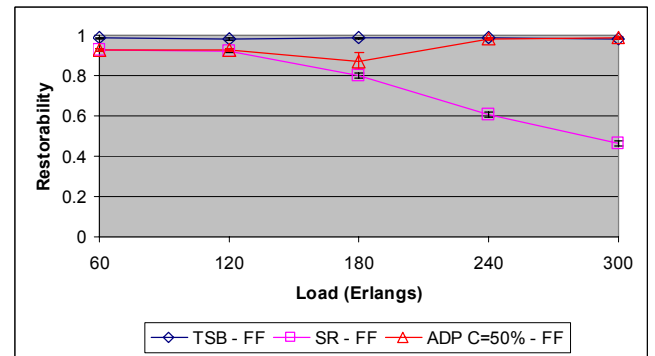


Fig. 7. Restorability.

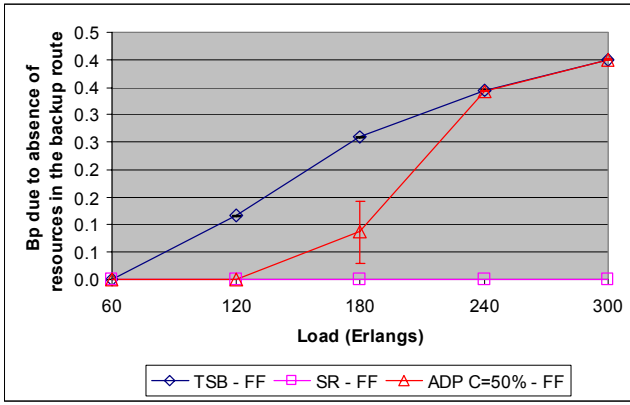


Fig. 8. Blocking probability (Bp) due to absence of resources in the backup route.

Figures 5 and 6 show that, as load increases, blocking probability also increases because network resources are scarce at higher loads for both survivability schemes. Figure 6 shows that due to allocation of redundant resource, the network utilization with TSB rises faster. This causes a blocking probability performance with TSB inferior to that achieved by SR (Fig.5). These results also show that the adaptive survivability scheme achieved a balanced performance between TSB and SR. When compared with TSB and SR, ADPS shows a more balanced increase in terms of blocking probability (Fig 5). This flexibility is achieved maintaining a level of restorability higher than 87% (Fig. 7).

Figures 9, 10, 11 and 12 show TSB, SR and ADPS (C=50%) performance related to blocking probability, average network utilization, restorability and blocking probability due to absence of resources in the backup route.

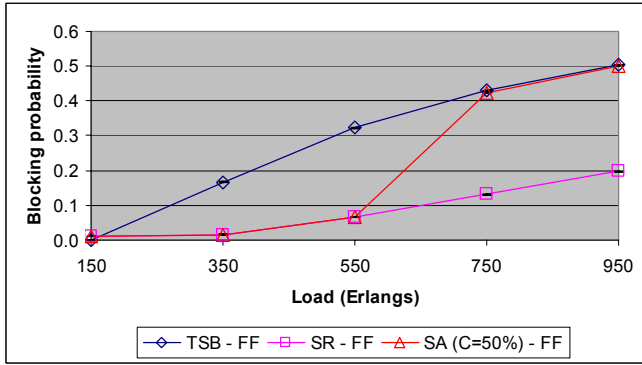


Fig. 9. Blocking probability.

The results obtained by applying TSB, SR and ADPS schemes under EON topology (blocking probability, average network utilization, restorability and blocking probability due to absence of resources in the backup route) show the same behavior when the Abilene network topology was used. It means that this observed behavior does not depend on the network topology, at least when comparing Abilene and EON topologies. Still analyzing the results achieved with these two important network topologies, we should notice that most part of the blocking probability of the TSB scheme occurs due to

absence of resource in the backup route. This happens because the TSB only establishes the connection if there are free resources on both routes (primary and backup). Nevertheless, all connection established have restorability very close to 100%.

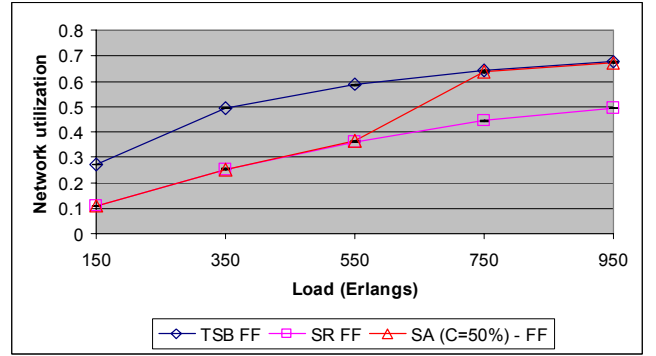


Fig. 10. Network utilization.

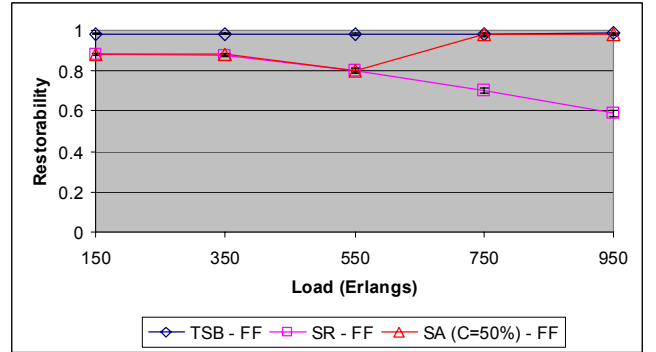


Fig. 11. Restorability.

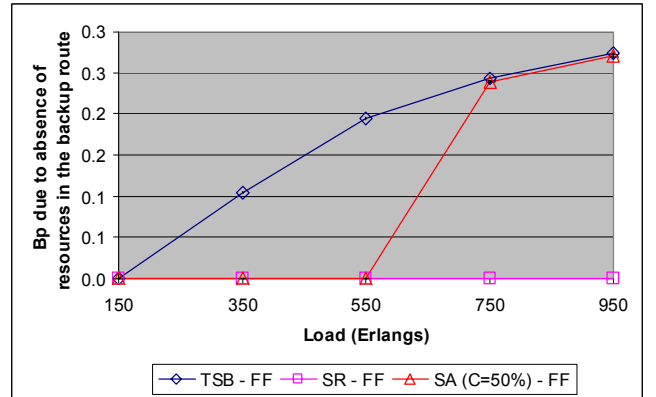


Fig. 12. Blocking probability (Bp) due to absence of resources in the backup route.

Figure 13 shows the blocking probability per source-destination pair (s,d) using ADPS C=50% and First-Fit wavelength assignment under traffic load equals to 300 Erlangs. The establishment of connections (s,d) occurs unequally. The spotted line in Figure 13 refers to the average blocking probability under 300 Erlangs.

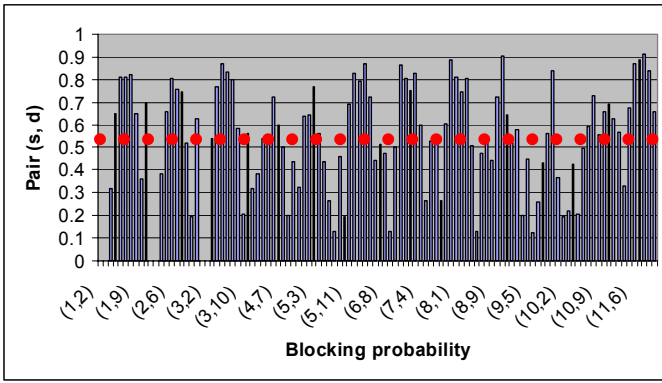


Fig. 13. Blocking probability per pair (source, destination) achieved with ADPS C=50% under 300 Erlangs.

One can notice that even under an uniform demand by the pairs of nodes (s,d), one group of pairs of nodes achieved blocking probability close to zero, while another group reached over 80% of blocking probability. Thus, an exclusive analysis of average blocking probability does not indicate whether the establishment of connections is fair. This unfair behavior is also noticed for all schemes studied under different traffic loads and for EON topology. However, it is important to point out that under either very low or high traffic loads there is a tendency of fairness related to blocking probability per pair of nodes (s,d). This is because the blocking probability is close to 0 (zero) or 1 (one).

Figure 14 presents the utilization per link using ADPS C=50% and First-Fit wavelength assignment under a traffic load equals to 300 Erlangs for Abilene topology.

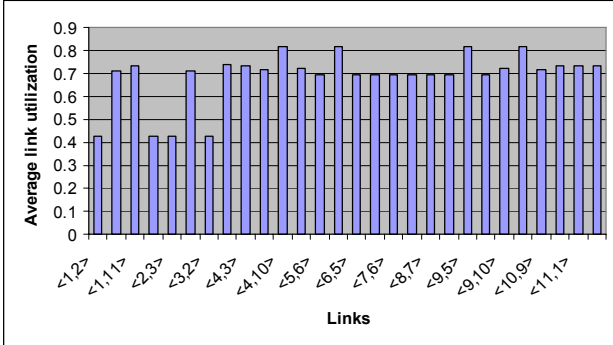


Fig. 14. Link utilization for ADPS C=50% with First-Fit under 300 Erlangs.

The link utilization is not uniform when the ADPS-FF C=50% is used (Fig. 16). This behavior is also observed for all schemes studied under different traffic loads and for EON topology. This non-uniform link utilization may suggest the use of a network with different numbers of wavelength per link. In terms of blocking probability per pair of nodes (s,d) and link utilization, ADPS achieved similar performance to that of TSB and SR.

We observe that ADPS presents a more flexible performance due to the adaptation capacity expressed by the parameter C. However, the parameter C has to be set up

previously. This adaptation enables ADPS to have the advantages of the TSB and the SR at the same time.

V. CONCLUSION

A new approach for performance evaluation of survivability services in all-optical network was proposed in this paper. It contains a set of measures that allow a more detailed evaluation of survivability strategies. It was also proposed a new all-optical survivability technique named Adaptive Survivability. This new technique was evaluated by using the set of measures suggested in this study. The Adaptive Survivability technique appears to have a flexible performance behavior taking into consideration its ability to behave sometimes as a protection strategy and other times as a restoration strategy.

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